

LECTURE NOTES ON EXPERIMENTS, EXPLORATIONS AND METHODS
(EEM) FOR DATA COLLECTION

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Explorations for Lecture 1 'Structure' and Lecture 2 'Phonology'

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Introduction

What follows here are examples of *possible prompts to experiments*—referred to here as **Explorations** (following the work of Tom Roeper, UMASS)—in showing how students engaged in linguistic courses which require *experimental design and data collection* can go about creating often simple ways (even ‘kitchen-table’ ways) of eliciting data from subjects. I organize the prompts for explorations in the following manner:

Section 1. In section 1.1, I present possible prompts to explorations as presented by various researchers such as Tom Roeper—who has come-up with the clever scheme of explorations, and a researcher I have personally known and collaborated with for several years on matters of child language acquisition—followed by other potential prompts suggestive of work done by researchers in the field, some of which are my own. This first section also provides some **theoretical background** to the potential exploration and could be read for hypothesizing aims and goals of the chosen experiment. Section 1.2 presents some background into Brain-related studies which involve Broca’s Aphasia (Grodzinsky). Section 1.3 includes Further Explorations and theoretical implications (Roeper). Included in this section is some discussion into Language Disorders (DELV test). Section 1.4 is an Overview of Child Language Acquisition (theoretical background).

Section 2. I leave the final section to what could be incorporated as possible prompts to explorations as found in the class readings of the text *The Psychology of Language* (Harley, 3rd). (I cite chapter and page number).

⇒ Symbol throughout indicates section for possible Exploration Topic.

[1] Subjects to investigate:

- (i) Young children (2-4-year-olds),
- (ii) Early school-age children (ages 5-9),
- (iii) Adult second language learners (L2/ESL-learners) (contrasting L1 and L2 results),
- (iv) American Sign Language subjects (ASL),
- (v) Learning disabilities subjects (e.g., Specific Language Impairment (SLI), Autism, Dyslexia, or subjects who have acquired cerebral lesions such as found in stroke patients (Broca’s aphasia)).

1. Lecture 1: Language Structure

These explorations in Section 1 correspond to Lecture-1 on 'Language Structure'.

1.1 Explorations (Roeper, 2007, 'Prism of Grammar' text).

Explorations ($\Rightarrow$), as presented here, are informal ways to glimpse into the subject's mental grammar. Regarding young children—and the nature of the grammatical errors they make—explorations allow us to explore maturational hypotheses behind the development of grammar, such as e.g., recursive movement, pronoun antecedent and binding, double question structures, functional-categorical and inflectional-morphology acquisition, etc.

For the other aforementioned subjects cited above, the same questions can easily apply, though perhaps with differing results. What one is always on the look-out for is the (sometimes lone) 'grammatical error'. Perfection of language, or so-called 'target grammar' of any subject is of little interest and use to us here—other than that it marks for a standard-native measure in order to contrast the *marked* ungrammaticality to that of the *unmarked* target grammar. While these explorations are informal (even 'kitchen-tablesque'), they can easily lead to formal experimentations in their own rights. Such explorations as used by both developmental linguists and psycholinguists often involve *storytelling, picture-sharing, and/or object-manipulation*. Even the very simple 'repeat-after-me' (word/phonology) test can reveal loads and loads of critical data of the sort which can give us a glimpse into the child's mental grammar and inner-language template. Such preliminary results allow questions and hypotheses to be developed, with the possibility of subsequent experimentation of a more formal setting to be devised. Most importantly, these explorations can be fun! Fun for both the subject as well as for the investigator. The creative aspects of designing such explorative experiments in of themselves make for a nice window into the type of questions linguists today are considering regarding the 'nature and design of language'. Let's begin...with double questions!

1.1.1. Double Questions

$\Rightarrow$ In a very simple explorative test, show a picture (or two pictures) of:

[2] (i) A boy (John) *catching fish with a fishing pole*

and

(ii) A girl (Mary) *catching crabs with a net...*

And then ask the child (this works to variant degrees by increment of age):

'Who is catching what how?

The correct answer relies on the child being able to map/parse double questions (where syntactic movement is involved)—to thematically map the correct ‘who-subject’ to the correct ‘what-action’ (and ‘how-manner’). Or the same test could be conducted at a gathering around the kitchen-table at dinner time: Ask a young child (2-3 years of age): Who is eating what? (Perhaps an older sibling is eating peas at this very moment, the mother chicken, the father drinking beer).

As it turns out, this test is not so simple for young children (3-4 years of age), as well as for older children (early school-age years) suffering from Specific Language Impairment (SLI). In fact, Roeper et al. have used this as a **communications disorder test**, and a **diagnosis for SLI**, called **DELV** (Diagnostic Evaluation of Language Variation). Adult/competent speakers have tacit knowledge that allows them to match the correct ‘wh’-questions—i.e., to link the member of each set one to one.

1.1.2 How children build and ‘label phrases’: From ‘Merge to Move’ operations.

[3] a. Draw a picture of *boat inside a house*

b. Draw a picture of a *house on top of a boat*.

⇒ Then ask the subject (young child),

- (i) show me the ‘boathouse’ (= a kind of house)
- (ii) Show me the ‘houseboat’ (= a kind of boat)

It may be the case that when you ask the child to show you a picture of a ‘houseboat’ (a ‘kind of a boat’ vs. not a ‘kind of house’), they might only have means to guess %50-50 at either picture. The fact that linguistics **movement** (a Broca region development) is required in order for the Phrase to mark its Head—with the first word as modifier¹ (in English the Head word is to the right in compounds)—reinforces the hypothesis that syntactic movement is not fully acquired at early stages of child syntax. See discussion below.

Some background Theory to a ‘Labeling Account’ of a Phrase

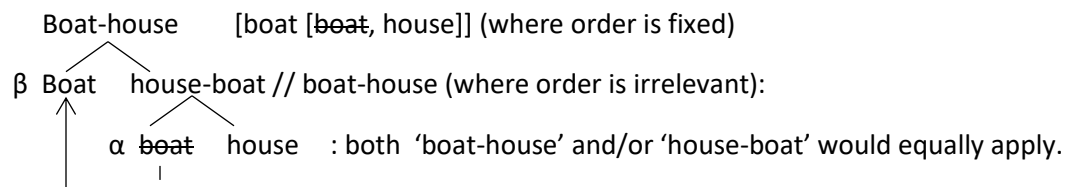
(See Galasso ‘Reflections on Syntax’ for a review).

A further requirement for movement may have to do with the **labeling** of a phrase. In terms of a ‘Merge-base’ theory of language acquisition, first-instance merge can only establish a primitive, unordered set of two items {a, b}—for example, an unordered {N, N}-compound of ‘boat-house’ would allow the ambiguous readings of either ‘a kind of house’ and/or ‘a kind of boat’ e.g., {*boat-*

¹ While English syntax is ‘Right-Branching’ (Head-initial) with the Head of XP positioned at the left of the complement, in morphology (e.g., compounding of [N+N] constructs), the Head of the compound is on the right.

house and/or *house-boat*}. In this first-instance of Merge, no *labeling* of the phase can yet be determined (as we are unsure which of the two Ns will be realized as the Head of the compound NP). This ‘unordered nature’ is the result of so-called ‘flat’ *sisterhood relations* where no hierarchy is yet established. It is only with second-instance of MOVE that order is derived out of a set {a {a, b}} which yields the proper recursive properties of hierarchical syntax—e.g., a 'house-boat' {house {house, boat}} now reads unambiguously only as a 'kind of boat'. It is this property of recursion that allows for projection and labeling of a phrase to take place (see [4] below):

[4] ‘boat-house’ (= a kind of house)



Theory-internal considerations are made within MP which stipulate for a **labeling algorithm** (LA) to scan syntactic objects (SO) and provide a label for one of the two SO-items (within the binary pair). In this case, MOVE is required in order for LA to break the flat sisterhood relation found of the SO. Move thus allows for one of the selected items to move up to a higher phrase, thus creating what is termed in MP as **dynamic antisymmetry**.

See ‘Reflections on Syntax’ papers below for labeling accounts:

https://www.academia.edu/43319709/Class_lectures_and_reflections_on_syntax_draft

https://www.academia.edu/42204216/Working_Papers_and_Reflections_on_Syntax_no_1_Lecture

⇒ A very interesting study regarding Head-initial vs Head final languages:

An experimental thought could be designed to see if there are so-called 'processing costs', coupled with Korean/Japanese (Left-branching, Head-final languages) vs English (Right-branching, Head Initial): viz., whether or not heavy NP shift would show processing delays (what ERP studies would show as P600s), given that head-final languages prefer long-phrasal strings before short phrasal strings (which mimic compositionality)--while English prefers short-phrases before long ones. in other words, regarding local first merge {X,Y} attempts at combining phases, in the first instance, kind of compounding of merged items which begins before compositionality of syntax takes over, we might expect English to show faster processing rates for NP-shifts e.g.,

Examples of possible prompts to experiment:

(1) 'John found [PP under his bed] [NP the book he was looking for the past year] (which shows non-canonical direct object NP-shift) while ,

2) 'John found [NP the book he was looking for the past year] [PP under his bed]' (canonical for English) but would be preferred by Korean speakers and not English speakers.

Example (1) shows NP Heavy shift moving lower down in the syntactic tree.

Recall, the for English XP, Head initial begins with just a single head, then proceeds with phrasal complement information which is substantially longer: So, English = [XP [H Head [Comp]]]: Hence, English should prefer short+long phrase structures (over long+short) in contrast to Korean/Japanese vice versa ([XP [Comp [Head]]]= Korean/Japanese.

⇒ Ask English vs Korean/Japanese ESL speakers which of the two sentence-types they prefer. They could give a rating scale of 1 thru 5 in terms of better to worse. Also mix up some ungrammatical structures to show some contrast, such as '*John under his bed found the book', * 'John the book found under his bed', etc,

1.1.3. The Determiner Feature [+ Definite]

'A' -> 'The' shift

⇒ Show a child a picture of two pens (a black one and a white one).

Now say to the child: 'Here's **a** white pen and **a** black pen'. (using the article 'a')

Now cover up the black pen and ask: 'Which pen did I cover up?'

Does the child say: '*The* black pen' or 'The black one' (using the +Def determiner 'the', or does she stay with what you said and simply repeat '*a* black pen or *a* black one'? e.g., 'You covered up 'A black one'...(Note that adult target grammar would use the +Def DP and say 'The black one', etc.

*(What you want to test here is to see if the young child can shift from the [-Def] determiner/article 'a' (found in the input) to [+Def] Determine 'the').

Young children may not have mastery of the [+definiteness] features associated with DP if the DP/Functional-category system is not fully operational at lexical stage-1.

- ⇒ Give a child two toy cars a blue one and a red one), and start to play with the red car. Hand it back to the child, (wait a few seconds), then ask:
- (i) 'can I have **the** car back'? (Does the child take the +Def 'The' to mean the 'red car. (that you just had)..or does the child just give *any* car back?
 - (ii) 'can I have **a** car back? (See what the child hands back: the blue or red one).

An interesting question is to see if there are other correlations with such a lack of DP and inflectional morphology (as shown next).

Some theory:

- (i) Lexical stage-1 (18-24 months) has been hypothesized as being operational without the functional categories DP, TP, CP.... Hence without DP features, Case, Inflection, Tense. (See Radford 1990, Galasso 2003, 2016).
- (ii) Functional stage-2 (24-36 months +) is seen as that stage where DP, TP, CP come on-line, triggered by Broca's regions.

1.1.4 Acquisition of INFlectional Morphology—Possessive {'s}

Here's a simple and fun way to see if a young child (2-3 years of age) has access to the inflectional possessive {'s}.

- ⇒ Point to an object X that belongs to Daddy and say to the child: 'That's daddy's X you can give it to me'.

Then point to another object X that the child knows belongs to the mother and say 'That's daddy's X.

How does the child respond? Does the child simply 'give the object'. If so, this may be a clue that the possessive inflectional (functional) feature is not being processed and that only the action verb 'give' in the phrase 'give it to me' overrides the functional features. Such early language void of inflections is associated with the lexical-thematic stage-1 (as shown in Radford 1990).

1.1.5 Acquisition of INFL: Tense, AGReement, Case: A Merge-only Lexical stage-1 (lack of Movement in stage-1) (See 'Overview' section below)

Let's consider what happens in a so-called **Lexical stage-1** (18-30 months) whereby such very young children lack all means for movement operations required of phrase labelling, tracking of movement out of embedded structures (similar to what we found of BA subjects), and syntactic agreement (Tense, Agreement, Case). Let's take each in turn.

Tense. Given that the regular formation of Tense is a Move-based operation (**affix-lowering** from TP down to VP), we would expect to see Tenseless structures at the lexical stage-1. Consider some examples of utterances below: (with adverbial ‘yesterday’ showing tense):

Past: yesterday, daddy go work. yesterday, mommy cook. Yesterday, me eat candy, etc.

Such tenseless bare-stem verbs are pervasive hallmarks of early stages of child syntax. (One need only look into the reams of literature of child syntax ranging from 2-3 years of age).

⇒ **Ask a child...Possible prompts for exploration:**

Have a child repeat after you a phrase with Tense (first regulars V + ed), and see if they can copy and repeat it: such as , ‘yesterday Mommy cooked pasta, Mary visited the zoo, John wanted milk’...Then do the same text with irregulars past tense verbs (which show no movement of affix lowering), such as Yesterday, I ate pasta, Mary went to the zoo, John drank milk’ and see if there’s any processing diction between: (i) the acquisition of regular Past Tense affix {ed} verbs, as well as any differences between Non-movement irregular Tense structures. Recall, that in linguistic theory, regular (decomposed) Stem + INFlectional affix formation triggers Movement (a Broca region processing) while irregulars (undecomposed) are stem-based without affix which triggers Wernicke’s region. The hypothesis is that developmental delays show up with regards to Broca Movement, and not with Wernicke root memorization. (This distinction forms the very basis of the Dual Mechanism Model, Dual Path-Way Model presented herein).

Broca’s Movement (affix movement):

a. [TP Mary [past: {ed}] [VP [[visit]ed] the zoo]] : [[stem] affix] : [[visit] ed]



Agreement. Subject-Verb AGREement is likewise Move-based, whereby Person and Number features of the Verb have to pair off (Check off) with its matrix DP noun. Consider:

[TP Mary [3P, Sg: {s}] [VP {Pres} drink -s] milk]] : [[drink]s]



(showing affix lowering of {s} onto the lower Verb):

‘Mary drinks milk. John plays ball. Daddy works. Mommy sleeps’ etc.

⇒ **Ask a child...Possible Prompts for explorations:**

The same type of 'repeat after me' test can apply also with AGR.

Have a child repeat: 'John plays ball' and see if they capture the verbal {s}, etc....

A classic lexical stage-1 (as found in Radford & Galasso (R&G) 1998) shows missing INFlection, Agreement, Tense (between the ages of 24 months till around 30-36 months of age).

Case. (Possessive).

Given that Possessive/Genitive Case is late acquired by children (See R&G), similar explorations could be devised in creative ways to see if young children have access to Case (either in the possessive form, or in the NOMinative 'I' form vs. the ACCusative 'Me' form....

Data (taken from R&G) show examples such as:

Possessive {'s}: Daddy sock (= daddy's sock), boy car (= boy's car), ...

My/His Possessive: Me car (= my car), Me kiki (= my cat), Him book (= his book), etc...

Nom v Acc Case: Me do it (= I do it), Him is big boy (= He is a big boy), etc.

⇒ Same Exploration. See if young children respond correctly to missing possessives. Ask a child to hand-over to you while saying <can you give me 'me car'? (my car) and see how they respond. (of course, a great way to collect data is to simply play with a young child and record spontaneous speech. This constitutes 'original data' collection).

1.2 Brain-related Studies.

Broca's Aphasia (BA) (Grodzinsky).

One example of how explorations are currently being used can be found amongst Broca-Aphasic (BA) subjects. Given that Broca's region of the brain is responsible for **syntactic-movement** operations, explorative designs can be devised to see if BA subjects can interpret movement at a distance (beyond just linear, local movement). The following explorations are pulled from Grodzinsky's work on BA subjects over several years:

Data on autism also show us how recursion (or the lack thereof) can be utilized as a diagnostic in terms of identifying Broca's aphasia (BA), where BA subjects infamously have a very difficult time handling movement operation, particularly those movements which span a distance over a

parsed utterance. Yosef Grodzinsky has spent a fair amount of his medical/linguistic career studying such cases. For example, the distance required of the recursive movement seems to matter in terms of processing, with closer move functioning much better than distant move.

⇒ Show pictures and action schemes to BA subjects in order to solicit movement-based data. Such schemes would be easy to devise and test (A Literature Review here of the data would also be most appropriate given BA subjects are inaccessible to most of us)

Consider BA data below when shown pictures with follow-up question 'Who's doing the pushing'.

(5) Which boy_t _____t pushed the girl? (Above Chance) : local move/surface S_VO order

(6) Which boy_t did the girl push _____t? (Chance): distant move /surface OSV_ order

BA subjects have difficulty with examples such as (8) above: (they only have 'chance' readings/interpretations when 'movement at a distance' is required of them). In example (5), the trace of 'boy' must coincide with the position which is further away sentence-final. Whereas the example in (6) the trace can be quickly recovered due to its close surface-level adjacency.

1.2.1 Young Children's lack of Syntactic Movement (passive to active inversion)

Other examples of movement which will illustrate below is how recursive movement develops over time (a so-called maturational theory of language acquisition, See Radford 1990).

For young children, (2-3-year-olds), they are often unable to follow and track movement operations as found in **passive to active conversions**.

⇒ For instance, one very simple exploration would be to ask a young child (even as old as 5 years of age) the following:

[7] First tell the child:

'Mary was kissed by John'.

Then ask:

'Who is doing the kissing?'

Young children often lack such syntactic movement and so can only process *linear and local overt language strings* (viz., ‘a beads-on-a-string’ theory in the manner of an input-output model (cf. **B.F. Skinner**). Such a surface strategy would force the child to interpret only that which appears in sequential order on the phonological surface) and so they say incorrectly ‘Mary’...that it is ‘Mary doing the kissing’ (instead of the correct answer ‘John’). Notice how the *passive* ‘Mary was kissed by John’ involves syntactic movement from the *active* (covert/underlying structure) ‘John kissed Mary’. (*Note: Carol Chomsky, **Noam Chomsky’s** late wife, was one of the first developmental linguists to publish such results). Today, the lack of movement found in young children can be accounted for by an underdeveloped Broca area in the front left hemisphere of the brain. This is similar to what we find of adult BA subjects. These are some of the ways ‘explorations’ can be utilized in psycholinguistic studies.

(I highlight the names *Skinner* and *Chomsky* here as the two square in fundamental ways in forming debates on so-called Dual Mechanism Models of language). See paper:

https://www.academia.edu/44274884/Notes_on_the_Biological_Basis_for_Language_The_Brain_The_Dual_Mechanism_Model_and_the_Two_Stages_of_Early_Child_Syntax_1

1.3 Further Explorations and Theoretical Implications (texts pulled from Tom Roeper’s work over the years).

1.3.1 Possessives Explored

In a series of explorations by various students and colleagues we began to pursue the question experimentally. The first step is to invent a context where several options are available and equally plausible. The first was invented by Sarah Gentile (2003), who gave a child three pictures based on familiar Sesame Street characters, but no story was presented (adults were tested in the next study).

[8] The Acquisition of Recursion.

Test 1.

A. Picture of Cookie Monster

B. Picture of Cookie Monster and his sister

C. Picture of his sister

⇒ Ask the subject: “Can you show me Cookie Monster’s sister’s picture?”

The results showed that about one third of the 3- to 4-year-olds took the conjunctive reading (Cookie Monster's and sister's picture) and chose Picture B.

Test 2.

[9] Quadruple possessive question:

Create some scenes which show the following: For instance,

What color is Mika's friend's dog's ball? — What color is Mika's brother's friend's flower? —
What color is Sho's friend's dog's tail? —

Child Language stage without recursive possessives (only lexicalization):

Independently assembling examples of early possessives, from Galasso (1999), here is what we find:

[10] I want me bottle. Where me Q-car? That me car. Have me show. Me turn. Me cat. Me pen. (2;6–2;8) ii. No you train. It's you pen. It's you kite. It you house? (3;2) iii. I want to go in him house. Him bike is broken. It's him house. b. Lexical i. Mine banana. Mine bottle. Mine car. Mine apple. Mine pasta (2;4) ii. My car. (3x at 2;4) My pasta. I want my key. It is my t.v.

⇒ Ask a young (2-year-old) child to repeat the phrases (or show pictures): This is: 'his bike', the 'Boy's bike', the 'Girl's doll', 'My bike' and see if they get the possessive case right—or do they repeat: 'Him bike, Boy bike, Girl bike, Me bike, respectively' showing no possessive case?

Further Reading, see Roeper:

[file:///C:/Users/jgios/Downloads/The Acquisition of Recursion How Formalism Articul.pdf](file:///C:/Users/jgios/Downloads/The%20Acquisition%20of%20Recursion%20How%20Formalism%20Articul.pdf)

http://people.umass.edu/roeper/online_papers/recursion%20bio%202011.pdf

The child uses pronominal reference through discourse without introducing new contextual references successfully, but the second “one or the other but not both” has three reconstructions that seem to be baffling. Nonetheless, many sentences allow an immediate use of Context:

[11] Show a picture of a piece of cake, then ask the child:

Look! Do you want to eat some?

This context engages NP-ellipsis reconstruction where some => some cake. Discourse subordinates this move to context, however, if we hear:

[12] Show a picture of some nails, along with the prior picture shown of the piece of cake, and ask the child:

I have some nails. Do you want to eat some__?

Here we are forced to reconstruct the anti-pragmatic NP: eat some [nails] even though a cake is present.

In pilot work Deanna Moore gave 10 3yr-olds pictures of Grover playing baseball and BB playing basketball. And said "Grover is playing baseball. Is he playing basketball?" 90% said "no" although a contextual referent for "he" exists.

[13] "I just bought some cake. Look here are some nails. Do you want to eat some?" We still anti-pragmatically reconstruct eat some nails although further back in the discourse we had a pragmatically acceptable noun-phrase.

Note that it is not "the last NP", but the object argument of a verb that we reconstruct. If we said:

[14] "I found some nails on the floor. Do you want to eat some__? The answer continues to be nails, the argument of the verb find, and not eat some floor (which is buried in an adjunct PP). The pure context-cake choice, if made consistently, might suggest an undiagnosed disorder.

This kind of comprehension of ellipsis and binding is common in children's readers and could easily be where a child goes astray. [See Roeper (2007)]

1.3.2 Language Disorders: DELV Test

The DELV test developed by Roeper et al show how Broca processing (Basal Ganglia) is responsible for processes which required two or more matching requirements:

For instance, consider so-called 'wh-double // wh-expressions' which require pairing and exhaustivity:

Who ate what? Adult: [Dad ate an apple and baby ate a banana]

How did she play what? Adult: [she plays the piano with her hands and drums with her feet] 1450 children were involved and the results comparing normal and disordered children 4-9 yrs show persistent large deficits: Red =typical, Green =Disordered: Typical non-paired answers: "Dad ate banana" or "hands and feet".

Exploring the detailed acquisition path of all quantifiers, particularly those that involve distributive (every) as well as collective (all) readings is of primary importance for understanding disorders and the challenge of language in primary school. (e.g., Merchant and deVilliers (2006) show every is produced late.)

Further reading, See Roeper:

http://people.umass.edu/roeper/online_papers/Microscopic%20MinimalismBU08.pdf

The best example of this are the results of 'Double Questions' as found in Roeper (2007, p 178):

[15] Double Questions

=> In a very simple explorative test, show a picture (or two pictures) of:

A boy (John) *catching fish with a fishing pole*

and

A girl (Mary) *catching crabs with a net...*

And then ask the child (this works to variant degrees by increment of age):

'Who is catching what how?

The correct answer relies on the child being able to map/parse double questions (where syntactic movement is involved)—to map the correct 'who-subject' to the correct 'what-action' (and 'how-manner'). Or the same test could be conducted at a gathering around a dinner table: Ask a young child (2-3 years of age): Who is eating what?

1.4 OVERVIEW OF CHILD LANGUAGE SYNTAX

First, a Theoretical Background

The stages of child syntax can be said to follow a trajectory of (i) non-recursive operation (found at stage-1, at approx. MLU of 2.5 words—e.g., what Andrew Radford (1990) called a ‘lexical/thematic’ stage devoid of movement and inflection), and (ii) a recursive, INFLectional stage (just after three years of age), whereby INFL as well as partial mastery of other functional categories (such as Tense, AGREement, Case, and fixed Word Order) begin to come on-line. (Radford assumes a maturational hypothesis here regarding the onset of movement. It could be argued that such a maturational onset of recursion is connected to the latter development of Boca’s area). I have gone on to call this stage-1 a ‘merge-only’ stage devoid of movement. (See Galasso 2016).

The classic stage-1 examples of ‘Him do it’, ‘Why daddy crying?’ show the same type of flat, non-recursive Merge-projections as found in our L2 data above. Consider the structure of ‘Him do it’ below:

[16] [VP Him do it] (showing no Tense feature)

[17] [TP He [+T] [VP does [VP ~~Him do~~ it]]].

In [17], we show:

- (i) the pronoun ‘Him’ raising from VP to TP to receive nominative Case (He) (T hands-over the feature to assign nominative Case),
- (ii) the verb ‘do’ receiving the Tense INFL-morpheme {s}.

The example of ‘Why daddy crying?’ is quite similar to that flat, non-recursive surface-level strategy we found of L2 speakers above:

[18] [QP why daddy crying]?

rather than the recursive structure of:

[19] [QP Why is [TP daddy is crying why]]? (base-structure prior to movement).

Here, the young child’s question-formation strategy is to simply attach the Wh-word ‘why’ above the already formed VP, forming a recurrent, non-recursive structure (with auxiliary verb omission):

[20] [VP daddy crying] => [QP why daddy crying]?

Clearly, such a simple, prosaic *Merge* operation whereby lexical items get attached to preexisting structure lends credence to the notion that early child English (at roughly 2 years of age) is predominately a surface-level phenomenon which places a high degree of burden only on surface-level phonological processes (absent full syntactic *Move* operations). In this sense, phonology is understood as a quintessential surface-level processing, whereby frequency of input and adjacency carries perhaps the highest priority in any overriding *optimality theory* of phonology.

⇒ ***Then, Ask a Child...Grammatical Judgements***

One simple explorative test which could be devised is to ask for **grammatical judgments** from young (lexical stage-1) children.

Present all forms and ask the child which is best to worst in terms of being correct:

- a. Why daddy crying?
- b. Why crying daddy?
- c. Why is daddy crying?
- d. Why daddy is crying?
- e. Why is crying daddy?

Present all forms:

- a. Me do it.
 - b. Me it do. (looking at word order)
 - c. I do it.
-
- a. Him do it.
 - b. Him does it.
 - c. He do it.
 - d. He does it.

You get the idea. Such explorations can solicit grammatical judgements which provide a glimpse into the child's mental grammar at his/her relevant stage.

Consider the referential properties of the English Pronoun system found below.

Such explorative could easily tease out whether or not children (L1) and/or Adult (L2) have full access to Referential pronouns.

Referential properties of English Pronouns

- [21] a. John_i hurt himself_i d. *He_i said John_i feels pain.
 b. John_i hurt *him_i e. Me_i hurt me_i (child syntax)
 c. John_i said he_{i/j} feels pain. f. Him do it

(where the index (i) correlates and joins the two items indexed insofar that John_i & Himself_i are one in the same constituent: John = himself), referred to as an anaphoric/antecedent relation).

By showing a picture of 'John hurting himself' vs 'John hurting another boy', and then by asking which picture/scenario matches which sentence, we can see if **Binding** operations are fully at work. For instance, if sentence (b) incorrectly matches-up with the scenario of sentence (a), (either for young children or for adult ESL (L2) subjects), then we know trace & binding of pronouns are not fully operational.

For example, one might easily question what it is exactly that makes [21b] so wrong (where *John* and *him* could correlate). As for myself, I simply cannot envision what line of difficulty arises by *parsing* (processing) the utterances *John hurt him* as having the equivalent meaning *John hurt himself*. (What is it that possibly blocks the two from having the same interpretation? Clearly, at some basic 'functionalist' level (of communication), the two expressions easily mean the same thing. Therefore, it must be blocked by some level of 'formalism', some level of syntax).

It is striking that such constraints as found with binding theory, while non-functional in nature, tend to hold as 'universals' across the world's languages. So, what sort of 'universal' do we have here (and where might they come from) which so apparently fly in the face of mere intuitive communicative assessments? One might rather ask here whether such a universal constraint on binding is in fact addressing some other conditions placed on language which speak less to (iconic) functionality and more to the very (abstract) nature of syntactic processing itself.

2. Exploration taken from the text (*The Psychology of Language*, Harley 2008)

(Page Referenced in order of page-number sequence in the Harley Text)

The examples and ideas for explorations come from multiple sources: the class text (*The Psychology of Language*, Trevor Harley (3rd, 2008), (in order of appearance, with page number indication), Tom Roeper's classic book *Prism of Language* (2007, MIT Press), Galasso's *Recursive Syntax* (2019, LINCOM Studies in Theoretical Linguistics, 61), as well as examples taken from Peter Gordon (slips of the tongue), classic Spoonerisms, and other personal works and sources I have collected and often used in my lectures over the years.

2.1 page 5. Differences found between DERivational) vs INFlectional Morphology.

Brain-imaging studies and ERPs reveal that processing of Der vs INFL morphology is different in terms of storage and retrieval. Consider the two distinct morpheme/affixes at work in terms of how we use brackets [] to indicate processing.

In fMRI studies, the following results have been widely obtained, showing how derivational morphology processes as lexical word-building [] or sometimes referred to as undecomposed chunking [], while functional grammar which incorporates INFL show distinct (bracket) processes:

- a. [Table] => Lexical: Noun
- b. [Dance] => Lexical: Verb
- c. [Hot] => Lexical: Adjective....all lexical categories which show [] processing, with items undecomposed wholes.
- d. [Brother] => (bi-syllabic), Lexical: Noun
- e. [Teach-er] => (bi-morpheme): Lexical: Noun via DER morphology...

All the above triggering Wernicke's area.

a-e in such fMRI, ERP studies show Wernicke's area/Temporal lobe, a region associated with lexical word [] storage and retrieval. Recall, that such lexical processes is [+Frequency] sensitive due the processing in this area of the brain.

But note in (f) below how [[teacher]s] processes both temporal lobe (for lexical/derivational [teacher]) as well as frontal lobe/Broca (Front-Left-Hemisphere) (for the INFL plural affix {s}):

- f. [[teacher]s] Lexical stem [teachers] plus functional-category affix {s }} = [[teacher]s]

One exploration would be to ask a Spanish-speaking (ESL) subject what they think (what is their own intuition) about the underlying processing of what was the historic singular/plural determine paradigm of 'La vs Lo' ('Lo' (once a masculine determiner now since lost due to loss of regular determiner forms of 'Lo-Los' (and has been supplanted via *suppletion* by the pronoun 'it')):

[22]

	Singular	Plural	
{	La	Las	}
	[La]	[La [s]] e.g. La Nina, Las Ninas	
Regular			
{	*lo	Los	}
	el		
		*(since lost due to suppletion of 'lo' with 'el')	
		Irregular	

Noting how regular processes is rule-based, thus provoking (Broca's area)/FLH processing, as opposed to any irregular formation which provokes a memory-based register (Wernicke's area).

⇒ Ask Spanish subjects if they overtly 'realize' that the {s} in 'Las' is marking plural, in other words, if they are aware that they are processing {s} as a plural marker. (They will most likely report that they DO!). Now ask the same subject if the {s} in 'Los' (as in 'Los Angeles') similarly marks for the plural—if they equally overtly realize its function as a plural marker. (They will mostly report that they DO NOT!). This is an interesting observation: it indicates that what was originally the same INFL plural marker {s)—which has since been lost due to suppletion of Lo with El, yielding the irregular formation of (singular) El, vs (plural) Los—is now very distinct in processing, with [Los] rather being processed as a whole lexical chunk, as opposed to 'Las' as a decomposed stem [La] plus affix {s}: [La [s]] processing.

In sum, the processing of 'Las' (e.g., Las Niñas) vs. 'Los' (e.g., Los Angeles) reveals a split between regular/decomposed [[]] vs irregular undecomposed [] structures:

[[la]s] vs. [Los].

*Again, recall that this distinction matches what we know of the **regular vs irregular** processing distinctions, with regular rules such as N + s = plural vs irregular whole: e.g., Spanish regular Determiner [[la]s] vs irregular determiner [Los], English regular [book] > [[book]s] vs irregular [foot] > [feet], etc.

What is very interesting here is that the **full-list hypothesis** in fact calls for such a morphological model that connects all *stems* to their respective *affixes* whereby the entire morphological item (word) comes via the lexicon directly as a stored chunk. Hence, such a Full-lexical entry model would process all inflectional words as holistic chunks processed accordingly: Plural [books], Verbal [speaks], Progressive [speaking], etc. Such a **full-list model** would suggest a Single Mechanism Model (SMM)—as opposed to a Dual Mechanism Model (DMM)—of language processing much akin to bygone eras of Behaviorism (B.F. Skinner) as well as current (strong) Connectionist Models of language. This SMM vs DMM could be tested (as seen below) by showing how the two final {s}'s in *[speaks] vs [fix] /fɪks/ behave (*hypothesizing that 'speaks' would be processed [speaks] in accordance with the SMM, much in the same way as how [fɪks] would be processed. A DMM model predicts a SMM for {s} in [fɪks] but a DMM {s} in [[speak]s].

Consider how the SMM vs DMM debate unfolds below (see example [38-39]):

2.2 Page 6. The stem-{s} vs the morpheme-{s} as in 'Fix' vs 'Speaks'.

Consider the two {s} forms as presented within the paradigm of recursive domain (structural hierarchy) (also showing the same processing distinctions as found in phonological assimilation):

A theoretically naïve view—i.e., stipulating that the two apparently adjacent final-position sounds of /_ks/ as found in the two words 'fix' /fɪks/ versus 'speaks' /spi:ks/ surely must equally get processed similarly (they do NOT)—calls on us to reconsider something much more insidious going on in the underlying structure of morphophonology:

[23] 'Fix' /fɪks/ vs 'Speaks' regarding final position /_ks/



The differences in processing resulting in distinctions held both developmentally (as found in child language) as well as for Second Language (L2) errors of omission: with a lexical stage-1 of child language often deleting the final /_s/ inflectional affix but never deleting final /_s/ stem-based elements (with similar findings for L2).

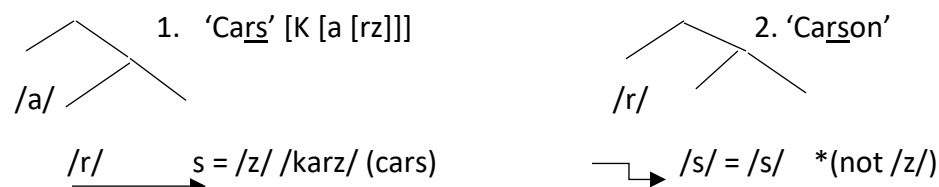
A note to consider: Chomsky never tires in explaining to linguistics students how what might appear on the **surface-level** (in the phonology) may not be an indication of what is actually happening at the **deep-level**, even hidden levels of language. ('Surface v deep' levels have been relabeled as Phonological Form (PF) and Logical Form (LF) respectively).

Assimilation in Phonology.

[24] Likewise, Consider the {s} in the environments: [_rs] vs [[_r] [s _]] (Text below taken from Galasso 2021).

Whether it be... constraints placed on *phonological assimilation* which stipulate that in order for the horizontal spreading of voicing to occur between two adjacent consonants, they must first be of a 'sisterhood' relation; whereby, for example, the /r/ in 'cars' provokes assimilation of plural /s/ => /z/, in contrast to the /r/ in 'Carson' which does not. (The former a structural 'sisterhood' relation, the latter a 'mother-daughter' relation in terms of hierarchy):

Carson' being broken-up by a *syllabic stress* /\$ /... /kar \$ sən/ (CVC \$ CVC).



Assimilation of 'voicing' applies between sisters. *No assimilation between mother-daughter.

2.3 Page 14. Lesion Studies / Dyslexia

[25] A cognitive neuropsychological literature review could be done here. For instance, regarding dyslexia, see references between:

- (i) 'Deep' dyslexia which affects semantics of word—such as when a subject is presented with the word CAT but reads DOG, or between, and...
- (ii) 'Surface' dyslexia which affects the phonological string, such as CAT becomes TAC,

The two types of dyslexia map onto the dual distinction in brain processing (The Dual Mechanism Model): Semantic-deep dyslexia showing effects as found in the temporal lobe region versus phonological (or syntactic) surface dyslexia showing Broca's effects. (See the works of Michael Posner and Marcus Raichle. A good summary is found in their book 'Images of Mind' (1994: Scientific America Library, New York, chapter 2).

Page: 14. Also, consider what's been said here regarding Broca's Aphasia (BA) (Grodzinsky). (Restate from § 1.2).

[26] One example of how explorations are currently being used can be found amongst Broca-Aphasic (BA) subjects. Given that Broca's region of the brain is responsible for **syntactic-movement** operations, explorative designs can be devised to see if BA subjects can interpret movement at a distance (beyond just linear, local movement). The following explorations are pulled from Grodzinsky's work on BA subjects over several years:

Data on autism also show us how recursion (or the lack thereof) can be utilized as a diagnostic in terms of identifying Broca's aphasia (BA), where BA subjects infamously have a very difficult time handling movement operation, particularly those movements which span a distance over a parsed utterance. Yosef Grodzinsky has spent a fair amount of his medical/linguistic career studying such cases. For example, the distance required of the recursive movement seems to matter in terms of processing, with closer move functioning much better than distant move.

⇒ Show pictures and action schemes to BA subjects in order to solicit movement-based data.
Such schemes would be easy to devise and test:

Consider BA data below when shown pictures with follow-up question 'Who's doing the pushing'.

(a) Which boy_t _____t pushed the girl? (Above Chance) : local move/surface S_VO order

(b) Which boy_t did the girl push _____t? (Chance): distant move /surface OSV_ order

BA subjects have difficulty with examples such as (b) above: (they only have 'chance' readings/interpretations when 'movement at a distance' is required of them). In example (b), the trace of 'boy' must coincide with the position which is further away sentence-final. Whereas the example in (a) the trace can be quickly recovered due to its close surface-level adjacency.

One nice topic here regarding a potential literature review section of a paper/project could be to summarize how brain imaging devices (e.g., ERP, fMRI, EEG, TMS) as well as brain signatures (such as N400, P300, P600) have contributed to our understanding of the brain-to-language corollary.

‘Center-embedding’ sentences.

Consider the two center-embedded sentence-types below (found on page 38):

- (c) Boris who is chasing vampires is sucking blood. (Q: Who is sucking blood?)
- (d) The vampire who is chasing Boris is sucking blood. (Q: Who is sucking blood?)

Given the above discussion that BA subjects have difficulty processing movement-based (trace-related) recursive properties of embeddings, it seems that they still get the readings correct despite their lack of syntactic movement—i.e., they use their ‘bottom-up semantic’ readings which disallows (c) ‘Boris is suckling blood’ *since people don’t suck blood* and rather employ the most obvious pragmatic/semantic-based reading that ‘*vampires suck blood*’—hence, 100% of the time they select the same reading for both sentences, seemingly going against the syntax in (c) and overextending the reading in (d) for both sentences.

This processes of allowing pragmatics/semantics to trump syntax is called ‘**Semantic Constraints**’, with **bottom-up** being relegated to pragmatics/semantics (word-meaning level) and **top-down** being relegated to syntax (sentence-level).

(See [31] below).

Page 21. Top-down v Bottom-up processing—‘Syntactic vs Semantic Constraints’

[27] One possible implication to so-called top-down/bottom-up processing is to see what happens to sentence-processing in so-called **Garden-path** sentences:

- (i) The horse [VP *raced* [PP past the barn]].
- (ii) The horse [[CP ~~that was~~ *raced* past the barn] Ø]
- (iii) The horse [[CP ~~that was~~ *raced* past the barn] *fell*] (garden path: P600 response?)
- (iv) The horse [[CP ~~that was~~ *raced* past the barn] [VP *fell* [PP onto the ground]]].
(no P600 response?)

In this sense, what we can say here is that there is some psycholinguistic contribution to language processing having to do with bottom-up v top-down processing. Here’s how it works:

In (i), the bottom-up reading is simple: this is an SVO sentence. (PP-Obj).

In (iii) bottom-up parsing is complicated by the deletion of the top-down CP (complementizer phrase ‘that-clause’). This allows a default bottom-up reading to overlap with (i), hence the extra verb ‘fell’ provokes a P600 signaling an illicit grammatical construction (despite the fact that it is grammatically correct—a garden-path sentence).

But in (iv), the added Prep Phrase (PP) forces the top-down processing of fell to be the main verb of the sentence (the horse falls), thus ‘repairing’ the garden-path error to a correctly processed structure. This repair is the result of top-down processing, where extra information (e.g., the PP-Obj percolates down to the bottom-up SVO and repairs the correct parsing of the sentences.

Experiments can be done regarding grammatical judgment tasks:

[28] Possible ‘Grammatical Judgment’ experiment:

=> Ask a subject sentence (iii), and see if it provokes a garden-path result (it’s wrong to them)?

Then read to them sentence (iv) with additional top-down ‘syntactic’ material and see if the reading elicits the same ‘garden-path’ response. If there is a difference in reading, in grammatical judgment (i.e., no garden-path response), this could be a result of top-down processing—whereby the extra syntax of a PP ‘onto the ground’ forces the verb ‘fell’ to be correctly read as the main verb, and not the first verb ‘race’. This is a classic case of syntactic constraints working to reinforce grammatical judgments of otherwise ambiguous sentences. (See page 38 for syntactically ambiguous sentences).

One last note here is to see how adjacency helps to determine structure. E.g., since race is the closest verb to the subject ‘The horse’, we are too quick to assume that this verb ‘race’ is the main predicate verb, when in fact the further away verb ‘fell’ is the closest verb based on **closeness in terms of structure**, and not **closeness in terms of surface adjacency**.

The nature of this type of repair to parsing is often termed a ‘constraint’: **‘Syntactic constraints on processing’**. I like to think of it as a sort of scaffolding which allows a speaker to look down into the sentence from above (top-down) and to end-up with a correctly parsed structure. This is the nature of top-down processing.

Note: Syntax is quintessential Top-down, while Semantic constraints (word-based, lexical retrieval) would be bottom-up. Recall, that an N400 (brain signature error) would be lexical/semantic based, while a P600 would be syntactic in nature.

Insert Note: Consider how Chomsky deals with ‘closeness of structure’ vs. Adjacency’ phenomena in his famous ‘can eagles that fly swim?’ phrase (below):

[29] **A look at Recursion** (See **page 39**).

The Four-sentences.

Opening remarks on Recursion:

The best way to see Recursion at work in language is to study the properties of embedded clauses. Below is my restatement of the **4-sentences**, each of which display unique properties of recursion.

1. Can eagle that fly swim? => Ask your subject: Who are we asking that they can do what?
2. Him falled me down! => (what is the nature or the error 'falled')?
 - a. If it is a 'flat' error, then it is processed as [falled]
 - b. If it is a recursive error, then it is processed as [[fall]ed]...

3. The horse raced past the barn fell. (garden-path sentence)

Note how the complete sentence cannot be as if a 'beads-on-a-string' theory, i.e., as a flat [] structure, but must be a recursive [[]] structure:

- a. [The horse raced past the barn fell]=> this gives rise to a garden-path false reading.
- b. [The horse [raced past the barn] fell] => this gives rise to correct/embedded reading.
 - (i) [The horse [~~that was~~ raced past the barn] fell]

4. I wonder what [that [is ~~what~~]] up there?

- a. I wonder what *[that ['s ~~what~~]] up there? (ungrammatical)
 - ⇒ Ask your subject if (a) is a 'well-formed' English sentence? Do they ever say that?
 - This is an interesting observation that in fact we don't say (b), despite the overwhelming reinforcing data which would otherwise suggest it to be correct, i.e., the simple fact that we say <That's> all the time.

Discussion: Speak to the two types of analyses (**Flat vs Recursion**). How can we argue for recursion in (b) being supported by linguistic theory over the flat structure in (a)?

For a complete discussion of the '*Four Sentences*' see my paper:

https://www.academia.edu/37650895/Four_Sentences_Opening_remarks_for_Ling_417_Child_Language_Acquisition_Fall_2018

[30] Let's consider what a flat (non-recursive) structure would look like for sentence 1:

v1 v2

a. [Can eagles that fly swim?]

So, if we are simply scanning strings via a process which only adheres to the 'adjacency-factors' of the string, then we should interpret that we are asking 'can eagles fly?' But let's consider now what sentence-1 looks like under a recursive structure:

b. [_x Can eagles [_y that fly _y] swim _x]

Now, if we consider the nature of recursive structures (as found with embedded strings), then we can see that indeed the closest verb to the subject [Eagles _x] (found within the _x constituency, or unit of structure), is in fact [swim _x] and not [fly _y]. As Chomsky puts it, it rather seems that it is due to some unique design of our human brain (a brain which gives rise to language) that allows us to instantiate immediately upon recognition (an innate recognition) the underlying recursive structure of [[]] over a flat structure []. This recognition is knowledge not learned in school, nor is it taught to us by our parents at an early age, but rather, comes for 'free' out of the *human design* of language.

Also, as addressed in my paper 'What makes language interesting For Ben' notice how closeness in terms of structure (deep level) overrides closeness in terms of adjacency (surface level):

- (i) [Ben is riding his unicycle] => a flat structure where adjacency works)
- (ii) [Ben who is my friend's son is riding his unicycle]

Now add Aux 'do' inversion to make question:

- (iii) [Is Ben who _ my friend's son is riding his unicycle?] (wrong flat structure).
⇒ If read as flat structure, processing is wrong!
- (iv) [Is Ben [who is my friend's son] _riding his unicycle]?
⇒ If read w. recursive embedding processing is correct)

(with [_] showing movement of Aux "do' inversion).

[https://www.academia.edu/15151583/Some notes on what makes language interesting For Ben Opening Remarks for Ling 417 Ling 329 Child Language Acquisition Spring 2014](https://www.academia.edu/15151583/Some_notes_on_what_makes_language_interesting_For_Ben_Opening_Remarks_for_Ling_417_Ling_329_Child_Language_Acquisition_Spring_2014)

[31] **Bottom-up constraints** may deal with **semantics**. For instance, in children (or Broca Aphasia subjects) for whom passive, embedded movement which requires a trace for distant references is a problem), note how only top-down/*syntactic* (distant movement) affects the correct interpretation vs bottom-up/**semantic** (local movement) as found in syntactic passive where semantics provides no clue):

a. The man the lion chased__ was eaten. Ask the Question: Who was eaten?

(‘man’ is correctly answered: above chance correct answer)

b. The lion the man chased__ was eaten. Ask the same question: Who was eaten?

(‘man’ again is answered but wrongly due to interference with semantics)

▪ Lions eat men, men don’t eat lions.

c. The ball was kicked by Mary (who kicked: ‘Mary’ correct answer, above chance) due to semantics : (people kick, balls don’t kick...)

d. Mary was kicked by the ball (who kicked: ‘Mary’ incorrect due to semantics).

e. Mary was kissed by John. Who is doing the kissing?

i. (Above chance incorrect answer: ‘Mary’) since semantics can’t help children or BA subjects recover the move-based/traced structure.

[32] (p. 21, p. 296) **Right Association**. ‘Bottom-up’ syntactic constraints also suggests that new information always gets processed at the lower levels in the syntactic tree and makes its way up (a kind of ‘syntactic percolation’ up the syntactic tree) in order to affect processing. For instance, consider ‘garden-path’ sentences here, and how new bottom-up info might help with interpretation/readings (what is termed ‘right association’. p. 297) since the new info is added right of the tree diagram, at the lowest level):

a. The horse [VP raced...] (VP = lowest level)

b. The horse [CP that [raced]].... (CP = lowest level) (CP= Complementizer Phrase)

c. The horse [VP-1 raced] past the barn *[VP-2 fell]

= garden-path (wrongly assume that [VP-1 raced] is the lowest level (main verb))

d. The horse raced past the barn fell [PP onto the ground],

See how the extra lower-level info Prep Phrase (PP) ‘onto the ground’ helps with interpreting the verb ‘fell’ as the true main verb. This is an example of bottom-up processing.

*(CPs introduce ‘that-embedded’ clauses in which the matrix embedded verb is not the main verb)

[33] p. 23. Does language use Rules?

One easy example to illustrate rules in language is to seek out so-called **default** rules. For example, the default plural rule in German is plural {s} (like English) although it is not the most frequently used affix (with the affix {en} more frequently used). Yet, German default for plural seems to be {s}. English too has default settings. For the past tense, it is the affix {ed} although there are other irregular options (e.g., sound change 'keep' > 'kept' /i/+ /ε/, etc.).

For English plural, the {s} is the default rule.

⇒ Here's an exploration you can do.

Ask a child if they can pluralize a nonce word/Noun such as {spligoot} and see how they would pluralize it. Recall, there may be two options:

- (i) Since the second part of 'spligoot' rhymes with /_goot/ (foot) they could say /spli_geet/ (goot= geet, as in 'foot' => 'feet').

So, see how the child (or adult) selects which plural affix:

Ask them: there is one 'spligoot' now there are two ____

What do they say? If the default is used, it would be: 'two 'spigoots'

Showing rule [[spigoot]s]. Recursion [[]] usually determines the rule.

*See Berko's 'Wugs Test' p. 145.

Finally, consider how peoples say the past tense to 'text'...Note how verbs which end in /__ t/ have two paths:

- (i) default {ed}, (yesterday, I 'texted'), [[text]ed]

- (ii) or irregular [[text]Ø] (no change: e.g., set, set, hurt, hurt).

There was once a time (when the word 'text' was first employed as a verb) that both options were in fashion: e.g., one could say: 'yesterday I texted', or, 'yesterday I text' (both versions were in competition with each other, but eventually the default version 'texted' won).

This **dual pathway** is implicated in brain processing:

- (i) [Stem [+ affix]] Rules = Front Left Hemisphere/Broca.
- (ii) [Stem + affix] whole word/stems memorized irregulars = Temporal lobe

*Note: This idea of two pathways will be called the **Dual Mechanism Model, Dual-Pathway Model'**: See **page 71** 'Recent models of how language is related to the brain', and **page 191** 'Morphology—processing complex words'.

Other examples are found in the English data, such as 'last at bat, he flyed-out right-field' (baseball), or 'She "out-Sally-Rided" Sally Ride' (Nasa astronaut), or 'There are two diffrent 'micky-mouses' (etc.). (fly => flew, ride => rode, mouse => mice... all which are irregular/SMM non-default formations). (*Noting that all irregular verbs/nouns take-on a SMM processing whereby the entire word/item entry comes directly via the lexicon as a chunk).

[34] p. 33 Hierarchical Structure of Syllables. Phonemic vs Syllabic stages of development.

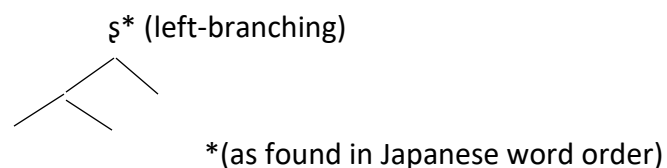
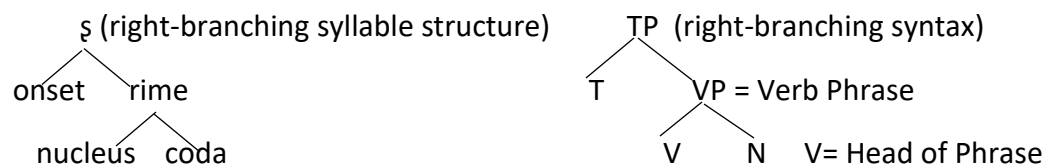
We now know that just like syntax, syllable structure is hierarchical in a nature, and presents like a syntactic tree (see below):

[Onset [rime: [nucleus, code]]] (see p. 33)

We know that for English hierarchy, the branching is rightward (where the Head of the Phrase is in initial position [VP [V, N]] (Not left-branching like Japanese). So too is it for syllable structure:

⇒ Exploration: older children/adults should clap-out the simple CVC word 'cat' as three taps: [C, V, C] = /k/, /æ/, /t/...but if the young children are not yet **phonemic** but rather are only **syllabic**, (an earlier stage of phonemic development) then they will clap twice, but only at the onset /k/ and rime /æ t/ whereby the rime is structured as one syllable (with right-branching). Note that children very rarely would tap twice as /kæ/ + /t/... which would show left-branching,

Branching Tree Structures (§) syllable structure vs syntactic structure (Tense Phrase)



⇒ **Tapping Experiments**: If young children are not yet at the phonemic stage (where they can separate individual phonemes), then they are at the so-called **syllabic stage** (an earlier stage). See where they tap: for example, is the word 'cat' tapped out as /k/ + /æt/, or /kæ/ + /t/? (Try to measure where the tap and phoneme map onto each other). (This is like how a first-grader points out the letters of the alphabet: A, B, C, etc., with their finger pointing to each letter, one at a time. Notice how early first-grades will tap out 'LMNO' as one single tap, with one finger point—this suggests that they have broadly incorporated

the four letters as a single holistic string-chunk [LMNO]). For young children at the broader syllabic stage, such chunking is what's going on in their processing, where a [vowel + consonant] (VC) structure will be processed as a single holistic sound chunk. Japanese ESL students likewise may fall into a more *broad* syllabic processing as they will be unable to separate a V from a C. This is because Japanese is *syllabic language* and not *phonemic language*. Once English-speaking children reach the more *narrow* phonemic stage, they will be able to tap, separate each individual phoneme within words. (First, second-grade teachers even have a document issued from the State Education Board which allows them to track students' individual 'phonemic-awareness' assessments and benchmarks).

*Right-branching syllabic tapping (to match right-branching syntax) is most likely what the child will produce—namely m children will tap out the structure for cat' as [k [æ, t]] (with /æ, t/ as one tap) and not [kæ [t]].

(See **page 243** 'Phonological Awareness').

[35] p. 36 **Functional Words**

Words like Determiners (A, The, This, That, etc.) and Auxiliary (helping verbs) (Do-Be-Have) are abstract in nature (closed class) and present as part of Broca's area processing. If maturation is a factor in child language development (see chapter 3), then Broca-related elements of language should be delayed.

Exploration: See if you can gather/collect original data (or gather a review of others' data) and see if young children I fact deleted Determiners and Aux Verbs in their utterances.

(See Radford & Galasso 1998 on the delayed usage of the functional possessive structure).

[36] p. 39 **Recursion**

Exploration: See if very young children have access to recursion (hierarchy);

Show them a picture of a 'house-boat' (a kind of boat) and a second picture of a 'boat-house' (a kind of house) the ask the child to show you e.g., a 'house-boat' and see if they just guess or get it right. If they guess %50-50, then you know that the child is not yet using recursive structure. In English, **compounds** are headed by the 'rights-side word' e.g., [house [boat]] is a kind of 'boat' (not a kind of 'house') etc.

*(See Tom Roeper section 1.1 herein this document)

[37] **p. 41 Passive to Active Transformations.**

Transformations are a result of syntactic moment, an abstract processing which is associated with Broca's area. So, again, like Functional words in [35] above) if abstraction/movement is delayed in young children, then they should have difficulties with 'active to passive' (and vice versa) transformations:

Explorations: ask a young child (or ESL adult) to move from deep structure/passive to surface structure/active in the following:

(i) (a) The ghost was chased by the vampire (**p. 40**)

(b) The vampire____(have them fill in the blank)

(ii) (a) The boy was kissed by John.

(b) John____

Carol Chomsky found that young children do have trouble with such active-passive movement.

⇒ Ask a first-grade child:

Tell them the sentence 'Mary was kissed by John' and then ask them 'Who is doing the kissing? You will find that they mostly say that it is 'Mary' doing the kissing (and not the correct subject of the action 'John'), etc, etc.

ESL adults may have similar problems with such movement-based operations (such as Functional words 'Determiner/Aux verbs' as well as syntactic movement of active-passive).

[38] **p. 71 (see also p. 191) Recent Models of how language is related to the brain.**

A Literature Review. Slips of the Tongue/Rat-eater Experiment

I'd suggest at this exploration to simply do a Literature Review of what the data reveal regarding a dual-pathway for language—i.e., how language is partitioned into two fundamentally different areas of the brain. How do the two areas exactly partition language?

See Broca's area and the work of Grodzinsky's **Brocas Aphasia**.

See Wernicke's area and the work leading to **Wernicke's Aphasia**.

These two regions of the human brain support the following language-processing distinctions:

- (i) Broca (front-left hemisphere/FLH): Inflectional morphology, Functional Categories, Movement operations,
- (ii) Wernicke (FLH/Temporal Lobe): Derivational morphology, Lexical Categories.

Functional Categories include: Determiners (A, The, This, Each, Some), Plural {s}, Tense {s}, {ed}, Auxiliary Verbs (Do, Be, Have), Complementizers ('That' which introduce embedded That-clauses), Case (I v Me, He vs Him), Word Order (Head-initial vs Head-final), etc.

This split model have suggested the **Dual Mechanism Model (DMM)**, the **Dual-Pathway Model**, as well as so-called *Race Models* (e.g., see Baayen).

One other point to make here is that if a Maturational Hypothesis is correct regarding the full operational onset of Broca's area for the child, and if it is correct that Broca is responsible for the more abstract, movement-based properties of language (e.g., Functional Categories, Recursive movement such as embedded clauses, active-to-passive and affix-level movement as found in morphosyntax, etc.) then a nice Literature Review should be able to reveal data which suggest such a maturational hypothesis—i.e., a stage where there is no inflectional morphology, no movement, and a lack of functional categories. (See Radford 1990, Felix 1992 (p. 111), and Radford & Galasso 1998).

[39] Ullman's D/P Model (Declarative/Procedural Model)

Similar to what we find above in [38], Ullman has proposed his D/P-Model (which map onto the same Broca/Wernicke distinctions).

⇒ Exploration: Given that INFLECTIONAL affixes should be separated (in processing) from Lexical stems (as suggested in the DMM above):

The linguists Peter Gordon has gathered thousands of 'Slips of the tongue' and has shown how such slips (mistakes) are not at all 'random' errors, but rather how the nature of such errors follows a 'systematic theory' which supports the DMM. Consider some slips below:

- (i) 'The car was driving'. (Target = intended utterance)
- (ii) 'The drive was caring' (=Slip)
- (iii) @ 'The driving was car' (@unattested, error never made)

Let's take a closer look at what the structure is here by doing a [lexical [functional]] analysis (given that we know [stem+ [affix]] sequences in 'Head-initial' languages like English follow this order).

Be V ing

(ia) The [car] was [driv [ing]] (= Progressive grammar: Be+V+ing)

(iia) The [drive] was [car [ing]] (= slip of lexical items N, V)

(iiia) The [driving] was [car] ____ (@unattested: not found in data)

If the stem V (drive) and the inflectional affix {ing} are indeed separated in processing, then they should NOT be able to move together. This is indeed what we find in the slip—they word 'driving' doesn't move in one fell swoop (given that 'driving' is not processed as a holistic chunk *[driving] but rather as a dual pathway, DMM processed [driv [ing]]. Only the two lexical stems can be exchanged in the slip (the Noun [car] and the Verb [drive]).

Note: For a strong **Connectionist Model** (behaviorism of the B.F. Skinner persuasion), all stem+affix formations would fall into a **Full-listing Hypothesis** (see pg. 191)—for e.g., vdrive would have contain the list 'vdrive' as the verb vroot), present tense [drives], progressive [driving]. Past participle [driven]...

For this **Single Mechanism Model** (SMM) all words along with their appropriate affixes have been to a large degree memorized as holistic chunks and are stored and retrieved in the mental lexicon (Wernicke's area/Temporal Lobe) as a single item []. This SMM goes against a Chomskyan view related to a Dual Mechanism Model (supported by the likes of Pinker and Ulman).

Note: See **page 191** 'Morphological Processing Complex Words'

Note: Rat-eater Experiment.

The linguists Peter Gordon came up with a clever experiment called the 'Rat-eater' experiment which showed how young children knew not to insert/keep inflections within compounds:

⇒ Ask a child: 'What do you call a person who eats *rats*?'

[[rat]s] a DMM processing

They will respond [rat-eater] and not [[rat]s]-[eater]. (They delete the INFL {s})

But when asked: What do you call a person who eats *mice*?

[mice] a SMM processing

They will respond [[mice]-[eater]] since mice is an irregular []-formation govern by the SMM.

It seems that children have tacit knowledge that in order to cement one large word out of two smaller chunks required for compounding—i.e., in order to string two nouns together [rat] + [eater] in forming ‘rat-eater’ (with the noun ‘eater’ itself formed via a derivational morphological processing of eat + er)—that the processing of the two lexical-category nouns (which form in the temporal lobe of the brain/Wernicke’s area) cannot incorporate an inflectional {s} since any functional-category inflection (arising from Broca’s area, a different area of the brain not responsible for word-building) would intervene in blocking the N+N lexical word-building process. Children subconsciously know to drop the {s} in rats yielding rat-eater. Hence, this experiment in of itself proves the DMM. If children processed all word/items as in the full-list hypothesis consistent of the SMM, then there should be no access to delete any part/affix of the word, given that for the SMM ‘rats’ would be formed as [rats], (all is *super-glued* together). We would wrongly expect rats-eater. (But there are no rats-eaters’ in the English-speaking universe). One could try this with Spanish, *comi-ratas*, but unfortunately, due to Spanish sometimes ‘left-branching’ word order, the inflectional {s} in ‘*ratas*’ always presents at the end and is not therefor stranded in an embedded position between two lexical items.

Here’s a fun kitchen-table experiment:

- ⇒ Ask a child: what do you call a person who catches dogs?
- ⇒ What do you call a person who bags groceries?

(Create your own examples here).

Do you find that the child responds: ‘dog-catcher’ (and not ‘*dogs-catcher’), or grocery-bagger (and not *groceries-bagger)?

In other words, it is as if children are being guided by their innate psychological processing which governs the following recourse to morphological sequencing (abiding by regional brain processes):

[Lexical + Lexical (compounding) + derivational] + inflectional]] (in that order):

- (i) With Lexical, Irregular, compounding, Derivational... processing being governed by Wernicke’s area /Temporal lobe, a []-process,
- (ii) With Functional, Regular, affix, Inflectional... processing being governed by Broca’s area/Front-Left-hemisphere, a [[]]-process.

The two processes can’t mix inside of one another, and when they do, as in some ‘exceptional-irregular’ compounds, these forms strike us as being wrong: consider how such inflect-embedded compounds sound to your ears: e.g., *mothers-in-law*, *attorneys-general*, (even the verb-

compound ‘*opened-fire*’). People often report that they want to ‘over-correct’ them to *mother-in-laws*, *open-fired*, etc. where the INFL affix is rightly at the end. These are exceptions which abide by different linguistic factors at work).

As a final note: This calls my attention to my own experiment I found by chance while speaking to my son after school one day. I asked: ‘What about taco tonight_s’? (noting the slip of the tongue ‘tonights’ whereby the plural {s} should have attached to [[taco]s], instead of being postponed to the adverb [[tonight]s], thus the error/slip. (Also note how the adverb [tonight] in fact does permit a functional inflection, such as possessive {‘s} [[tonight]’s] party, or contracted {is}, tonight’s going to be fun!) In any case, the mere fact that the displacement took place in the slip of the tongue dispels any notion that the plural noun ‘tacos’ would be processed as a SMM [tacos], given that the {s} was removable and displaced within the error. This shows how the word ‘tacos’ is rather *decomposed* (broken-up) [[taco]s] and not undecomposed [tacos] (as might be suggestive of a full-list hypothesis where there are no need for morphological rules. In the rat-eater experiment above we saw that regular plural [[rat]s] was decomposed while irregular plural [mice] was undecomposed.

One would be hard pressed to find errors which can occur wild and randomly, without a basis in theory. For instance, just imagine the {t} in ‘taco’ being similarly displaced:

- (i) What about [_acos] tonight t? (where the first consonant /t/ in ‘taco’ gets displaced and postponed to the end of the sentence, just like what occurred with the plural {s}.
- (ii) Or even, try to image the two adjacent letters/phonemes simply being exchanged:

e.g., What about tacot sonight?

Of course, such ‘wild’ errors never occur because they can’t! Our linguistic theory disallows such wild random errors. For once thing, in the above examples, it is the brain that tears apart the lexical /t/ (in [tonight]) and the functional {s} in [[taco]s]. Brain processing doesn’t function as a single conveyer-built, but rather there are two fundamentally different areas at play here, as supported in the *dual mechanism conveyer built* (smile!).

Aside from Visual Dyslexia (whereby a child may read ‘tac’ for ‘cat’, quite common occurrences in young readers), there are essentially two types of Phonological Dyslexia which seem to mimic this DMM: Reading Dyslexia can be broken up into three forms: (surface dyslexia (word-based) and deep dyslexia (semantic-based), and phonological dyslexia (phoneme-based)).

One type involves **Phonological-Dyslexia** (PDs) of the type which seems to support only surface/visual memorization of words as chunks, and not phonological Rules). PDs have a hard time pronouncing words using phonological rules (e.g., they can’t ‘sound things out’). Whereas **Visual-Dyslexics** (VDs) may read the two words ‘have’ and ‘wave’ as having the same rhyming vowel /æ/ (since they *look* alike visually on the surface), PDs have problems reading non-words such as ‘sleeb’ (since this requires a ‘sound-it-out’ strategy based on ‘phonological rules’). This interesting characteristic of PDs’ inability to read easily pronounceable non-words like ‘Sleeb’ while being able to read the verb ‘Sleep’ shows their inability to map phonological units to speech while their lexical memory of words learned is preserved.

- (i) **Surface-Dyslexics** (SDs) have a hard time with irregular words such as ‘broad’ vs ‘brode’. However, SDs do seem to have their phonological rules intact—e.g., they can read without difficulty regular words and non-words alike with equal success. This suggests that while SDs can use phonological rules it is only their direct route to the lexical item itself which has been impaired: for SDs: It’s a lexical-route storage & retrieval problem, not a phonological problem.

The DMM, Dual-pathway model suggest that such SD’s only have an indirect, non-lexical route—i.e., the impairment regards the lexical retrieval route with the impairment involving the Wernicke/Temporal Lobe, and Left-posterior occipital lobe. This is most obvious with their inability to read irregular words, words which must be memorized via a direct lexical route.

- (ii) **Deep-Dyslexia (Semantic Dyslexia)** seems to be an impairment at the deeper semantic level of reading whereby a word such as ‘symphony’ might be read as ‘orchestra’ (similar in meaning, but not in phonology). Clearly, DSs have an impairment at the smaller phonemic-level of speech representation. They are not decoding on a phonological level, but their lexical storage & retrieval processing is working at an exclusive semantic level.

⇒ **Exploration:** I’d suggest a summary or literature review of how these two types of dyslexia map onto the dual-pathway, Dual Mechanism Model.

[41] p. **294 Parsing**

Parsing in the psychological process of stringing a sentence together in order for it to make real-world sense. One interesting idea is that Syntax can help with Semantics.

Consider the following syntactic structures (involving word order) and see how just a slight adjustment in syntactic word order can change meaning:

(i) Boris loaded the truck with hay **p. 311**

(ii) Boris loaded hay onto the truck

(iii) John painted the house red.

(iv) John painted red on the house,

Clearly example (i) suggest that Boris complete loaded the truck with hay while (ii) suggests that just some hay was loaded onto the truck.

Similar with (iii, iv): 'John painted the house red' (with the adjective 'red' at the end of the sentence) shows that 'red' complements/modifies the 'whole house' (John's house is red. It is a red house). (The noun 'red' does not serve to modify the whole house, but rather the verb 'to paint' (to paint the color-red)).

⇒ **Exploration:** See if young children or even adult ESL speakers can tease out the difference between the two. Create some similar sentences and see if your subjects can understand the differences or do they suspect that they have the same meaning? Your results might suggest that young children as well as adult ESL have trouble with deeper syntactic processing and may only be processing/parsing at a **surface/shallow level**.

[42] **One very interesting note about ESL Shallow-Processing/Parsing**

Harald Clahsen and Claudia Felser have shown that ESL subjects have a parsing strategy that only relies on the surface/shallow level (as oppose to a deeper syntactic level which involves movement).

For instance, ESL subjects often operate on a surface-shallow level when operating with Auxiliary Verbs to form questions: For instance, they may simply tac on a Wh-word to an already parsed, pre-existing (S)entence structure, such as: 'What she does in the morning'? Or may provide double tense structures such as 'Did she visited him already'?

Such *shallow* parsing/processing would have the following 'tac-on' structure:

[wh-Q What [S she does in the morning]]? [Aux Did [S she visited him already]]?

- ⇒ **Exploration.** See if ESL and or young children respond to such shallow-processed sentences are ungrammatical. If not, one might argue that they view them as being grammatical correct based on their limited parsing mechanism (which only allows shallow processing)

See Clahsen, H., & Felser, C. (2006). Continuity and shallow structures in language processing. *Applied Psycholinguistics*, 27, 107–126.

[43] Ambiguous readings and how syntax can help with semantics.

One last famous example of how syntax can help with semantics is to see how movement operations can help to disentangle ambiguous meanings:

Consider the famous sentence below:

- (i) John saw Mary with a telescope.

This sentence is ambiguous: Who has the telescope? Is it John who sees Mary via the telescope (i.e., John is holding the telescope), or is it Mary who is holding the telescope? The two sentences as is are ambiguous (we can't get to the semantics). But notice how Movement (as syntactic operation) helps with semantics: If we move the PP (prepositional phrase 'with a telescope' to the front of the sentence, then we know that the PP is serving in an adverbial role to describe John:

With a telescope, John saw Mary.

= How did John see Mary? ... With a telescope.

Hence the PP 'with a telescope' must be coming out of the DP-subject 'John' syntactic node and not from the DP-object 'Mary' node.

Note how the second reading where 'Mary has the telescope', such movement is not allowed:

*With a telescope, John saw Mary. (≠ Mary with a telescope).

- (i) [[John saw with a telescope] Mary. (With a telescope, John saw Mary).
(ii) [John saw [Mary with a telescope]].



The complete tree structure in (i) would have the PP branching out of the VP since the VP holds as the DP-subject 'John' as its constituent. The structure (ii) would conversely hold the PP as branching out of the DP-object 'Mary' since 'Mary' and 'with the telescope' are of the same constituency.

Some links to further Readings:

[https://www.academia.edu/44166191/The Y Model A Synopsis of Lectures in Linguistics Reading List and Topics on Language Structure Phonology and Syntax](https://www.academia.edu/44166191/The_Y_Model_A_Synopsis_of_Lectures_in_Linguistics_Reading_List_and_Topics_on_Language_Structure_Phonology_and_Syntax)

[https://www.academia.edu/42275126/The Myth of Function defines Form as the Null Biological Adaptive Process and the Counter Linguistics based Response Accumulative Lecture and Topics for Research](https://www.academia.edu/42275126/The_Myth_of_Function_defines_Form_as_the_Null_Biological_Adaptive_Process_and_the_Counter_Linguistics_based_Response_Accumulative_Lecture_and_Topics_for_Research)

[https://www.academia.edu/43319709/Class lectures and reflections on syntax draft](https://www.academia.edu/43319709/Class_lectures_and_reflections_on_syntax_draft)

[https://www.academia.edu/42204216/Working Papers and Reflections on Syntax no 1 Lecture](https://www.academia.edu/42204216/Working_Papers_and_Reflections_on_Syntax_no_1_Lecture)

[https://www.academia.edu/44274884/Notes on the Biological Basis for Language The Brain The Dual Mechanism Model and the Two Stages of Early Child Syntax 1](https://www.academia.edu/44274884/Notes_on_the_Biological_Basis_for_Language_The_Brain_The_Dual_Mechanism_Model_and_the_Two_Stages_of_Early_Child_Syntax_1)

[44] **Word Recognition, Reading, Dyslexia** (For 'Reading Non-words', see pg. 212).

Consider the non-words KNNOW (know) vs. FLLICK (flick). The first is a non-irregular word, the second is more regular in formation. Ask the question: which word is faster for you to process? Findings show that KNNOW processes faster than FLLICK. If so, this suggests that vision (irregular) may in fact process faster than auditory phonological-regular words...

Exploration: Ask a native English-speaking person which word is faster to process? Findings show that KNNOW processes faster than FLLICK. It seems vision is tied to irregular-spelling formation while regular-spelling formations are more inclined to being phonologically produced. This is a simple test to tease out the cited dual-route processing for 'word recognition' as well as 'reading' (see observations for reading and dyslexia below).

For Dyslexia/Reading Problems (surface, deep), see pg. 220.

- (i) 'Surface Dyslexia' (SD): Impairment with irregular 'Top-down' (surface-visual words e.g., 'steak') which has an irregular phonology since 'steak' should sound like 'speak' /spik/, but rather sounds like 'lake' /lek/. People with SD have impairment with the memory-visual irregular dual-route of word-processing. SD subjects highly depend on bottom-up auditory-phonological cues.
- (ii) Phonological Dyslexia (PD): impairment with regular "bottom-up" phonological processing such that DP subjects are unable to sound-out even highly regular 'bottom-up' words, e.g., SLEEB, which should have a direct phonological accessing route due to regular SLEEP, can't be pronounced. (It seems likely that some visual interference may be at work here as well). PD subjects highly depend on top-down visual cues.
- (iii) Deep Dyslexia (DD): Impairment where the semantics (not then phonology) seems to gain direct access to word selection/processing: e.g., when DD subjects read the word ORCHESTRA, they say the word SYMPHONONY, etc.

It's interesting to type-cast these three Reading Problems into two categories: (1) Top-down/Vertical (SD), vs (2) Bottom-up/Horizontal (PD), since vertical relies on memorization, visual frequency, even semantic knowledge (DD).

*A Final Note on Dyslexia 'TAC for CAT': It's worthwhile to note here one such popular sequence which manifests the above *dual-route* processing of vertical 'Top-down' vs horizontal Bottom-up' methods of processing. Let's consider the two major components of dyslexia, (i) semantic (deep dyslexia) vs (ii) phonemic (surface dyslexia).

First, let's assume—as we suggest above regarding speed of word recognition differences found between (irregular) KKNOW and (regular) FLICK—that vision does play a role in this reading-impairment (we'll call it 'Reading Impairment-A' (RI-A)). Let's follow the sequence of processing steps below, as the child with RI-A tries to read the word CAT:

- (i) First, the child uses vision as a processing mode to read [CAT]: this is an unsegmented, idiomatic treatment of the visual word as a chunk (typical of what we find in very early school-age speech whereby a child might process the letters/picture [LMNO] as an unsegmented, idiomatic chunk—where the student points to only one of the four letters when asked to show what he means by LMNO (one sound/letter brings along the other three for free). This is an unsegmented visual cue of the sounds LMNO, all strung together as an [idiomatic]-chunk.
- (ii) Now the child want to read and say CAT: Once the child processes visually the word [CAT], it becomes recognized and engaged in a mapping of 'sound /kæt/ to meaning' e.g., [/kæt/ = {furry animal with whiskers}].

- (iii) But now, the next processing step is to move the visual mode (which is unsegmented at this point) to a properly segmented phonemic representation whereby individual letters [CAT] map onto phonemes /k, æ, t/. (In other words, vision hands-off to phonology).
- (iv) But here is where it goes all wrong for the RI-A subject: as if building letter-blocks (bottom-up), where bottom block should be /k/, top block /t/, the process of moving and shifting vision to sound gets reversed, and the subject goes top-down and utters the top block first /t/ and /k/ last, uttering the word TAC for CAT.
- (v) It seems that this form of reversal/dyslexia is a byproduct of some form of entanglement in processing between (i) the visual mode of word recognition and (ii) mapping of idiomatic recognition from vision to individual phonemic speech. Certainly, the child here knows what a CAT is (the semantics is intact), it's just the mapping from (top-down) sight to (bottom-up) sound which gets mired in a processing interference.

This is one such example, albeit perhaps just a bit over-simplified for illustrative purposes, of how the dual route processing might account for differing aspects of dyslexia.

When we speak of 'processing', recall that all this means is that there is some tacit, inner-mode of computation happening with the processing of the external input, such that a young child, say, at a CV: gemination stage of syllabic development would hear the external output of /kɪti/ (a true CVCV structure where each phoneme is independent) as CV:CV (duplicated) /kiki/ (a kind of auto-segmental processing whereby the first sound /k/ is duplicated twice in the consonant slots, and the last sound /i/ is duplicated twice for both vowel slots). External /kɪti/ becomes Internal /kiki/. This is a result of 'processing' on a syllabic developmental level.

Of course, we already know the same dual route goes far to explain the cut between Wernicke's vs Broca's aphasia, or even neurological degenerative disorders such Alzheimer's vs Parkinson's disease (where the former is an X=X/Memory vertical route, and the latter an X+Y=Z/Rule horizontal route).

Syllabus to this class can be found at:

https://www.academia.edu/44797367/Linguistics_438_538_PSYCHOLINGUISTICS_Spring_2021_On_Line_teach_syllabus_Alternative_Modes_of_instruction_AMI_On_line_Teach

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